

For example, if $X = \{1, 2, 3, 4\}$ and $Y = \{2, 5, 8, 3, 9\}$, then

$$X - Y = \{1, 4\} \quad \text{and} \quad Y - X = \{5, 8, 9\}$$

d) **Unary Set operator ‘Complement’:** X' denotes ‘complement of X ’.

‘Complement of a set X ’ is a special case of the ‘Difference of Two Sets’. For defining ‘complement of a set’, we need to know the value of the Universal set U , where universal set is the largest set under consideration in the particular context. If U denotes Universal set under consideration, then complement of X , denoted by X' , is defined as $U - X$.

If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $X = \{2, 5, 8, 3, 9\}$, then

$$X' = \text{Complement of } X = \{1, 4, 6, 7, 10\}.$$

Different notations for the complement of the set A include (i) A' (ii) $\neg A$ and (iii) A^c .



Fig. 1.2: If area shaded in red on L.H. S. represents A , then area Shaded in red on R. H. S. represents Complement Set of A .

1.2.7.3 More Clarifications

- i) **operands/ argument/ inputs; value/ result/ output:** Let us understand these concepts through examples.

In one of the examples above, we mentioned if $X = \{1, 2, 3, 4\}$ and $Y = \{2, 3, 5, 8, 9\}$, then $X \cup Y = \{1, 2, 3, 4, 5, 8, 9\}$.

The values $\{1, 2, 3, 4\}$ and $\{2, 3, 5, 8, 9\}$, which are **given as inputs** to the operation of union are generally called **operands, or arguments, or even just inputs**. The value $\{1, 2, 3, 4, 5, 8, 9\}$ obtained after application of the operation of union is called **value, or result, or output**.

The above definitions, discussed just now, are explained for operations. These definitions are equally applicable to relations also. For example, we mentioned above that $7 < 5$ returns *FALSE*. Then the numbers 7 and 5 which are **given as inputs** to the relation of ‘less-than’ are generally called **operands, or arguments, or even just inputs**. The returned value, viz. *FALSE* after application of the relation ‘ $<$ ’ is called **value, or result, or output**.

- ii) **operation vs. operator**

The two concepts of ‘operation’ and ‘operator’ are quite similar and are often confused for each other. For our purpose, the **symbol** for union, namely $\cup$, is called the **operator**. On the other hand, if we are more interested in

getting $\{1, 2, 3, 4, 5, 8, 9\}$ from the operands $\{1, 2, 3, 4\}$ and $\{2, 3, 5, 8, 9\}$ then we talk of **Operation of union**. Similarly, in the case of **relations**, the **notation/symbol** <used for 'less than' is called **operator**. On the other hand, if we are more interested in getting *FALSE* from the operands 7 and 5 then we talk of **relation of 'less than'**.

Check Your Progress 4

1) Let $X = \{1, \{2, 3\}, \{4, 5\}, 6\}$ and $Y = \{1, \{2, 4\}, \{5, 6, 7\}\}$. Find the following:

a) $X \cup Y$

b) $X \cap Y$

c) $X - Y$

d) $Y - X$

.....

.....

.....

.....

1.3 NUMBERS: FROM NATURAL TO COMPLEX

1.3.1 Some Standard Number Sets

We just mention some frequently occurring number sets and their standard names & notations.

$\mathbf{N} = \{1, 2, 3, \dots\}$ is called the set of **natural/ counting numbers**

$\mathbf{W} = \{0, 1, 2, 3, \dots\}$ is called the set of **whole** numbers

(*It may be noted that for quite some time now, even 0 is being treated as a counting/ natural number. Thus $N=W$, and hence, there is no need for the concept of whole number and for a separate notation W .)

$\mathbf{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is called the set of **integers**,

$\mathbf{Q} = \{x: x = p/q \text{ with } q \neq 0, \text{ and } p, q \in \mathbf{Z}\}$ is called the set of **rational** numbers.

Remarks: It can be easily seen that $\emptyset \subset \mathbf{N} \subset \mathbf{W} \subset \mathbf{Z} \subset \mathbf{Q}$.

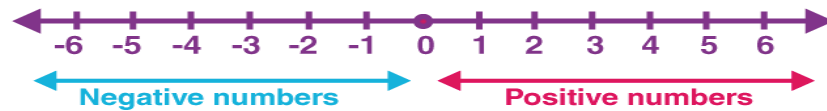
Except $\mathbf{Z} \subset \mathbf{Q}$, other set-inequalities given above are obvious. For showing $\mathbf{Z} \subset \mathbf{Q}$, let $n \in \mathbf{Z}$, then $n = (n/1) \in \mathbf{Q}$. Further, $3/4 \in \mathbf{Q}$, but $3/4 \notin \mathbf{Z}$

1.3.2 R, The Set of Real Numbers

Next, we briefly discuss the numbers, viz. *the real numbers*, which are encountered most frequently in our attempts at numerical problem solving.

The *Set of Real numbers* is denoted by $\mathbf{R}$ and has geometric connotation as explained below.

Take a straight line and mark two distinct points on it, naming the *left mark* as 0 and the *right mark* as 1. Then every point on the straight line (extended infinitely in both directions) can be made to correspond to a number, called a **real number**.



The correspondence to members of $\mathbf{N}$ can be obtained as follows: For obtaining a point corresponding to number 2, mark a point on the line on the *right side of 1* and at a distance equal to the distance between points, on the straight line, corresponding to 0 and 1. Similarly, points for 3, 4 etc. may be obtained.

To get points corresponding to **negative members** of $\mathbf{Z}$, let us consider how the point on line corresponding to (-1) is obtained. Mark a point to the *left* of 0 at a distance equal to the distance between points corresponding to 0 and 1. This point corresponds to -1 . Moving similarly more and more *to the left* yields successively points on line corresponding to $-2, -3$, and so on.

To get a point corresponding to a **rational number** say 3.45, divide into 100 parts the length of line between points corresponding to the points 3 and 4, and then move 45 parts to the right from the point corresponding to 3. The point on the line so obtained, is the point that corresponds to the rational number 3.45. Similarly, points corresponding to other *rational* numbers may be obtained.

Each of the numbers—so obtained by corresponding it to a point on a straight line in the manner described above—is called a **real number**.

The set of all such numbers is called **Set of Real Numbers**, and is denoted by $\mathbf{R}$, where $\mathbf{R} = \{x \mid x \text{ is a real number}\}$.

It can be easily seen that the *length of hypotenuse of a square* of length one unit is $\sqrt{2}$ units. Thus, moving on the straight line from 0 towards *right* for a distance of $\sqrt{2}$ units, yields a point, which corresponds to the real number $\sqrt{2}$. But it has been proved that there is *no rational number* p/q which equals to $\sqrt{2}$. Such a **real number, which is not a rational number**, is called an **irrational number**. It has been proved that number of irrational real numbers *is more than* number of the rational real numbers. The **set of irrational numbers may be denoted as $\mathbf{R-Q}$ or, as $\mathbf{R \sim Q}$** .

Remarks: From the discussion above, the containment relation ' $\subset$ ' between various number sets can be further extended to include $\mathbf{R}$ as follows

It can be easily seen that $\emptyset \subset \mathbf{N} \subset \mathbf{W} \subset \mathbf{Z} \subset \mathbf{Q} \subset \mathbf{R}$

The argument above establishes the inequality $\mathbf{Q} \subset \mathbf{R}$ (as, $\sqrt{2} \in \mathbf{Q}$, but $\sqrt{2} \notin \mathbf{R}$).

1.3.3 Real Number Intervals

Here are the definitions presented through examples:

- the interval $[3, 7.5]$ denotes the set of all real numbers between 3 and 7.5 and it *also includes* both the boundary numbers 3 and 7.5. Such an interval is called a **closed** interval.
- the interval $] 3, 7.5]$ (also, denoted as $(3, 7.5]$) denotes the set of all real numbers between 3 and 7.5 and it includes 7.5, but not 3. Such an interval is called a **semi-closed or semi-open** interval.
- the interval $[3, 7.5[$ (also, denoted as $[3, 7.5)$) denotes the set of all real numbers between 3 and 7.5 and it also includes 3, but not 7.5. Such an interval is also called a **semi-closed or semi-open** interval.
- the interval $]3, 7.5[$ (also, denoted as $(3, 7.5)$) denotes the set of all real numbers between 3 and 7.5 but, it includes neither of 3 and 7.5. Such an interval is called an **open** interval.)

1.3.4 Complex Numbers

Next, we discuss how the above sequence of inequalities of containment in number-sets is further extended to **C, the set of complex numbers**

The set of complex numbers developed in context of attempts at solving equations of the type $x^2 + 1 = 0$, for which no real number solution exists. A new type of number i was introduced for which $i^2 = -1$ was assumed. Then C is defined as $C = \{z: z = x + i y: x, y \in \mathbb{R}\}$

(Note: $x + i y$ is also written as $x + y i$)

Then inequality $\mathbb{R} \subset C$ follows from the following argument: Let $x \in \mathbb{R}$, then $x = x + i 0 \in C$. Thus, $\mathbb{R} \subseteq C$. Also, there are many complex numbers including, say,

$1 + i 2 \in C$ such that $1 + i 2 \notin \mathbb{R}$. (Also, alternatively, as discussed above, $i \notin \mathbb{R}$, but $i = 0 + 1 i \in C$). Hence, the proof. Also, using the earlier set inequalities, we get

$$\emptyset \subset \mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset C$$

Remarks: The notations $\emptyset, \mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, C$ are standard and can be used without giving their respective definitions/contents. However, the other set names like Sq_int are *not* standard; we could have used some other name instead of Sq_int . For all such set names, in every new context, its definition has to be given afresh.

1.3.5 Number Miscellany

- **Number**(Examples: 3, -12, 8.7, $2/3$, $\sqrt{2}$, π , and $7 - 3i$, where 3 is a natural number, -12 is an integer, each of 8.7 and $2/3$ is a rational number, each of $\sqrt{2}$ and π is an **irrational number**. Each of the numbers 3, -12, 8.7, $2/3$, $\sqrt{2}$,

π is also a real number. The number $7 - 3i$ is a complex number, but it is not a real number.

Each of the numbers quoted here is such that the number does not have the preceding properties. For example, $\sqrt{2}$ is an irrational number, but it is not a rational number, it is not an integer, and also it is not a natural number),

- **Arithmetic/Algebraic operations** (for example, $+$, $-$, $\times$, $\div$ are all **binary** operations, as each requires **two** numbers for its application.
- Examples of **unary** operations on numbers—which requires *only one* number for its application—include (i) **Exponent/power** e. g, x^3 ; and in general, x^n , for any fixed n ; and (ii) $|x|$ denoting **absolute value of a real number x** , defined as

$$|x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

For example, $|5| = 5$, and $|-5| = 5$

- **Arithmetic relations**(Examples: $=$, $\neq$, $\leq$ and $\geq$),
- **Fraction & Rational Number:** The number expressions $3/5$ and $6/10$ are two distinct/different fractions, but represent the same **rational number**. **Two fractions p/q and r/s** with $q \neq 0 \neq s$, are equal as rational number, if $p \times s = q \times r$, as mentioned just above that distinct fractions $3/5$ and $6/10$ equal as rational numbers. Now, let us look at **Prime Numbers**. A **natural number p** is said to be a **prime number**, if the only natural numbers which divide p are 1 and p itself. For example, each of the numbers 2, 3, 5, 19, and 31 etc. is a prime number. However, 4 and 21 etc are **not** prime numbers. **Though the natural number 1 has property somewhat like a prime number, but 1 is NOT considered as a prime number.** The reason is that the number 1 is a **unit** in N , i. e. for 1, we have $1 \times 1 = 1$. For any other natural number $q \neq 1$, there is no s in N , such that $q \times s = 1$.
- The symbol π , **spoken as Pi**, is standard notation which represents a real number defined by the relation $c = \pi d$, with c denoting circumference of a circle and d denoting the diameter of the same circle. **The real number π is irrational.** But the proof is beyond the scope of the course.
- **Nomenclatures for Number sets:** The symbol **N** is the initial letter of *Natural* of the full name ‘Natural Numbers’ for the set. The symbol **Z** is the initial letter of the German language word ‘Zahlen’ meaning ‘Integer’. This is why, *the set Z is sometimes also denoted by I*. The symbol **Q** is the initial letter of *Quotient* of the full name ‘Quotient Numbers’ for the set. The symbol **R** is the initial letter of the word *Real* of the full name ‘Real Numbers’ for the set. And finally, the symbol **C** is the initial letter of the word *Complex* for the full name ‘Complex Numbers’ for the set.

1.3.6 Properties of Real Numbers as Ordered Field

The following properties of real numbers—w.r.t the two binary operations ‘+’ and ‘×’, w.r.t binary relation ‘<’, and the unary relation |·|—are *used frequently*, knowingly and unknowingly, in calculations and proofs of statements involving real numbers.

- a) **R, the set of real numbers, satisfies the following properties (as a *Field*) w.r.t the two binary operations ‘+’ and ‘×’:**

For all x, y and $z \in \mathbb{R}$, we have

- 1) Closure properties: $x + y \in \mathbb{R}$, and $x \times y \in \mathbb{R}$,
- 2) Associative properties: $(x + y) + z = x + (y + z)$; $(x \times y) \times z = x \times (y \times z)$
- 3) Existence of (unique) identities: 0 and $1 \in \mathbb{R}$, such that $(x+0) = (0+x) = x$ and $(x \times 1) = (1 \times x) = x$
- 4) Existence of (unique) inverses: for $x \in \mathbb{R}$, $\exists$ (there exists) $(-x)$, such that $(x + (-x)) = ((-x) + x) = 0$, and for $x \neq 0, \exists (x^{-1})$, such that $x \times x^{-1} = x^{-1} \times x = 1$
- 5) Commutativity: $(x + y) = (y + x)$; and $(x \times y) = (y \times x)$.
- 6) Distributivity of ‘×’ over ‘+’: $x \times (y + z) = x \times y + x \times z$.

- b) **R, the set of real numbers, additionally satisfies the following properties (as an *Ordered Field*) w.r.t the two binary operations ‘+’ and ‘×’, and the binary relation ‘<’:**

For all x, y and $z \in \mathbb{R}$, we have

- 1) If $0 < x$, and $0 < y$, then $0 < x + y$; and $0 < x \times y$;
- 2) **Trichotomy property:** for $x \in \mathbb{R}$, exactly one of the following three is true (a) $x < 0$, (b) $x = 0$, and (c) $0 < x$.
- 3) **Transitivity property:** if $x < y$, and $y < z$, then $x < z$.
- 4) If $x < y$ then $x + z < y + z$, (monotone property of addition)
- 5) If $x < y$ and $0 < z$ then $x \times z < y \times z$, (monotone property of multiplication)
- 6) **Archimedean property:** given two positive numbers A and B , there is a natural number n such that $B < n \times A$

- c) **R, the set of real numbers, additionally satisfies the following properties (Metric/ norm) w.r.t the two binary operations ‘+’ and ‘×’, and the binary relation ‘<’, and unary operation $|x|$ (modulus)**

For all x, y and $z \in \mathbb{R}$, we have

- 1) $|x| > 0$ and $|x| = 0$ if and only if $x = 0$;
- 2) $|x| = |-x|$;
- 3) $-|x| \leq x \leq |x|$; and,
- 4) **Triangular inequality:** $|x + y| < |x| + |y|$.

Mod function introduces the distance function $\| \cdot \|: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$

such that, For all $x, y \in \mathbb{R}$

$$\|x - y\| = |x - y|.$$

Then, for all x, y and $z \in \mathbb{R}$, we have

- 1) $\|x - y\| > 0$ and $\|x - y\| = 0$ if and only if $x = y$
- 2) $\|x - y\| = \|y - x\|$, and
- 3) $\|x - z\| \leq \|x - y\| + \|y - z\|$. (Triangular Inequality)

1.4 DECIMAL REPRESENTATION OF RATIONAL AND IRRATIONAL NUMBERS

The method is explained through examples. First, we consider examples of decimal representations of *rational* numbers, and then those of *irrational* numbers.

- i) **The terminating decimal representation of rational:** Consider $3/8$. We know it is represented in decimal notation as **0.375**. This representation is **terminating**, as it contains only finite number of digits.
- ii) **Non-terminating recurring/ repeating decimal representation of rational numbers:** $7/3$ is represented in decimal notation as $2.333\dots$, in which the digit 3 continues repeating for ever, i. e. the decimal representation is **non-terminating**, but the sequence consists of a single digit 3, which repeats itself.

Similarly, the rational number $29/7 = 4.142857142857142857\dots$, in which the sequence 142857 of decimal digits repeats itself for ever or the sequence is **non-terminating, but repeating**.

Notation: Instead of repeating infinitely many times some digits in notation, the notation $\overline{142857}$ is used for $142857142857\dots$

Thus, $7/3 = 2.3333\dots = 2.\overline{3}$, and $29/7 = 4.\overline{142857}$.

- iii) **Decimal representation of an irrational number, which is always non-terminating non-repeating sequence of digits.** For example

$$\pi = 3.14159\dots, \text{ Similarly,}$$

$$\sqrt{2} = 1.4142857\dots,$$

where there is **no pattern** in ‘...’

The process of converting decimal numbers to fractions: Let us understand the process through the following example:

$$187.0145083 = \frac{1870145075}{100000000}.$$

On R.H.S., in the *denominator*, under the decimal point, we write 1. Then, for every digit (zero or non-zero) after the decimal *of the numerator*, in the denominator, a 0 is placed starting from the position on the right of 1.

Then, the expression on R.H.S. is simplified by dividing both numerator and denominator by their common factor. For example, first, we divide both numerator and denominator by 25 to get $= \frac{748058003}{400000}$. If there are more factors common to numerator and denominator, then we can further simplify the fraction through division by the common divisor.

1.5 REAL NUMBER EXPONENTIATION

We know that **the notation x^3 stands for $x \cdot x \cdot x$** , i. e. x multiplied by itself three times. In general, for $n \in \mathbb{N}$, **the notation x^n stands for $x \cdot x \dots (n \text{ times})$** , i.e., x multiplied by itself n times. Then x is called the **base** of x^n and n is called the **exponent** or **power** of x^n .

For example,

$$(19)^4 = 19 \times 19 \times 19 \times 19;$$

$$(-15)^3 = (-15) \cdot (-15) \cdot (-15)$$

$$(3/4)^5 = (3/4) \cdot (3/4) \cdot (3/4) \cdot (3/4) \cdot (3/4) = (3) \cdot (3) \cdot (3) \cdot (3) \cdot (3) / (4) \cdot (4) \cdot (4) \cdot (4) \cdot (4).$$

$$(4) \cdot (4)$$

$$= (3)^5 / (4)^5.$$

In general, $(x/y)^n = x^n/y^n$

The definition can be extended to zero and negative integer exponents as follows:

$x^0 = 1$, and $x^{-1} = (1/x)^1 = (1/x)$. In general, $x^{-n} = (x^{-1})^n = (1/x)^n$ for $n \in \mathbb{N}$.

For example, $(19)^{-3} = (1/19)^3 = (1/19) (1/19) (1/19)$.

The following rules for exponent can be easily verified for $m, n \in \mathbb{Z}$ (and, even, for $m, n \in \mathbb{R}$)

$$x^{m+n} = x^m \cdot x^n;$$

$$x^{m \cdot n} = (x^m)^n, \text{ where}$$

$(x^m)^n = (x^m) (x^m) (x^m) \dots$ repeated n times, if n is positive integer (or even 0). And,

$x^{m \cdot n} = (x^{-m})^{-n} = (x^{-m}) (x^{-m}) (x^{-m}) \dots$ repeated $(-n)$ times if n is negative.

$$x^{-n} = 1 / x^n, \text{ and } x^{m-n} = x^m / x^n;$$

For extending the definition to fractional exponents, let us first understand what $\sqrt[n]{2}$ means? It means the number such that $\sqrt[n]{2} \times \sqrt[n]{2} = 2$

Similarly, $\sqrt[4]{7}$ means $\sqrt[4]{7} \times \sqrt[4]{7} \times \sqrt[4]{7} \times \sqrt[4]{7} = 7$,

$$\text{Or } \sqrt[3]{(8/11)} \text{ means } \sqrt[3]{(8/11)} \times \sqrt[3]{(8/11)} \times \sqrt[3]{(8/11)} = 8/11.$$

The notation $\sqrt[n]{2}$ is used for $(2)^{1/2}$, and the notation $\sqrt[4]{7}$ is used for $(7)^{1/4}$.

The notation is in consonance with $\sqrt[3]{(8/11)} \times \sqrt[3]{(8/11)} \times \sqrt[3]{(8/11)} =$

$$(8/11)^{1/3} \times (8/11)^{1/3} \times (8/11)^{1/3} = ((8/11)^{1/3})^3 = (8/11)^{(1/3) \cdot 3} = (8/11)^1.$$

In general, for $n \neq 0$, $\sqrt[n]{x}$ denotes $(x)^{1/n}$, where x may be any real (or complex); and the number n , so far, is a natural number.

Further, for $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, $x^{m/n} = x^{m(1/n)} = \sqrt[n]{x^m}$.

and for $n \neq 0$, $x^{m/n} = (x^{1/n})^m = (\sqrt[n]{x})^m$.

So far, we discussed addition, subtraction etc. of only exponents. Next, briefly we discuss arithmetic **operations on bases**.

First, we discuss cases in which base is **either a product $(x \cdot y)$ or a quotient (x/y)** , then

$(x \times y)^a = (x^a \times y^a)$ and $(x/y)^a = (x^a / y^a)$ for any real or even complex number a

For addition and subtraction of x^m and y^n , the expressions $x^m + y^n$ and $x^m - y^n$ are treated as almost independent irreducible expressions, except when x and y have some common factor $\neq 1$, as explained by the following example:

$3^5 + 6^2$, we may write as $3^5 + 6^2 = 3^5 + (3 \times 2)^2 = 3^5 + (3 \times 2)^2 = 3^5 + (3)^2(2)^2 = (3)^2 [(3)^3 + (2)^2]$

Check Your Progress 5

- 1) Express the decimal number 18725.4625 as a fraction.

.....
.....

- 2) Express $31/12$ in decimal notation.

.....
.....

1.6 LET US SUM UP

In this unit, we had a sort of bird's eye view of the (most) fundamental concepts of Mathematics, viz. (i) the number, the basis of conventional mathematics; (ii) the *set*, the basis of modern mathematics. In this respect, we discussed the operations having following **operation-symbols** (operation-symbols are generally called **operators**):

- i) Set operators (or set operation-symbols): (examples) $X \cup Y$ (for union), $X \cap Y$ (for intersection), $X - Y$ (for difference), and X' (for complementation) etc.
- ii) Set-relational symbols: (examples) equality ($=$) of sets; subset ($\subseteq$); proper subset/ superset ($\subset$), and, not a subset of ($\not\subset$), belongs to ($\in$), does not belong to ($\notin$) etc.

- iii) Algebraic/arithmetic operators (or algebraic operation-symbols):
(examples) $+$, $-$, $\times$, $\div$, $\sqrt{x}$, $|x|$, $\log x$, e^x etc.
- iv) Algebraic/arithmetic relational symbols: (examples) $=$, $\neq$, $>$, $\geq$, $<$, $\leq$ etc.

1.7 KEY WORDS

- Cross-Product of sets** : Cross-product of sets for two non-empty sets A_1 and A_2 , denoted as $A_1 \times A_2$ is the set of all ordered pairs (a_1, a_2) , where $a_1 \in A_1$ and $a_2 \in A_2$. The concept may be generalized to any finite number of sets say $A_1, A_2, \dots, A_n$, denoted as $A_1 \times A_2 \times \dots \times A_n$. **The cross-product of n sets** is the set of all n -tuples of the form $(a_1, a_2, \dots, a_n)$, where $a_i \in A_i$ for $i = 1, 2, \dots, n$.
- Number** : It is used for counting, measuring or labelling. It is an abstract concept, which is made tangible through number-names, called numerals, e. g. *five* is purely an idea, which becomes tangible through the numeral 5.
- Operation on a set X** : It is quite similar to the concept of *relation*. An operation on X maps the cross-product of X with itself n times, where n is the arity of the operation. But it returns an element of X itself. For example, operation of *square* on N , the set of natural numbers, involves a single number, hence, it is an operation of arity one; the operation of *sum* involves two numbers, hence its arity is two. But in both the cases of *square* and of *sum*, the value returned is a single number.
- Relation on a set X** : It is a subset of cross-product of the X with itself n times, where n is the arity of the Relation. For example, *is-male* involves a single person, hence, it is a relation of arity one; *is-mother-of* involves two persons, hence its arity is two. A relation returns exactly one of the values 'true' or 'false'. For example, if Sheela is not mother of Mohan, then the value of the relational expression *is-mother-of* (*Sheela, Mohan*) returns false.
- Set** : Set is a well-defined collection of objects, where, a collection C is said to be well-defined, if we think of or pick up anything, say x , then we should be able to tell whether x belongs to C , or does not belong to C .